

Parasitology

Original MCQ Reviewer with Answer Key and Rationales

Concept coverage follows the uploaded Parasitology chapter. The question stems, clinical scenarios, distractors, and rationales are newly written for study use rather than copied or lightly paraphrased.

8.11 Parasitology

1. A clinician suspects pinworm infection in a child with nocturnal perianal itching. Which specimen collection method gives the most appropriate diagnostic material?

- A. Routine stool O&P; only
- B. Serial cellophane tape preparations
- C. Urine sediment
- D. Sputum examination

2. Which stool finding may form after eosinophils disintegrate during an inflammatory response to parasites?

- A. Charcot-Leyden crystals
- B. Howell-Jolly bodies
- C. Hemosiderin granules
- D. Schuffner dots

3. A patient acquires a tapeworm after eating undercooked freshwater fish. Which parasite best fits this route?

- A. *Diphyllobothrium latum*
- B. *Taenia saginata*
- C. *Hymenolepis nana*
- D. *Echinococcus granulosus*

4. African trypanosomiasis caused by *Trypanosoma brucei* is transmitted by which vector?

- A. Tsetse fly
- B. Sand fly
- C. Mosquito
- D. Tick

5. A stool specimen from a patient with prolonged diarrhea contains coccidian oocysts that stain with a modified acid-fast method. Which organism is compatible with this finding?

- A. *Cystoisospora belli*
- B. *Dientamoeba fragilis*
- C. *Entamoeba gingivalis*
- D. *Trichomonas vaginalis*

6. Which specimen would be least useful when looking for *Trichomonas vaginalis*?

- A. Vaginal discharge
- B. Urethral discharge
- C. Urine
- D. Feces

7. Which permanent staining method is especially useful for identifying *Dientamoeba fragilis* in stool?

- A. Trichrome stain
- B. Modified acid-fast stain
- C. Gram stain
- D. India ink

8. Which intestinal protozoan is strongly associated with waterborne outbreaks after ingestion of infective cysts?

- A. *Giardia lamblia*
- B. *Entamoeba gingivalis*
- C. *Balantidium coli*
- D. *Trichomonas vaginalis*

9. An amoeba recovered from a gingival lesion contains ingested leukocytes. Which organism is most consistent?

- A. *Entamoeba gingivalis*
- B. *Entamoeba histolytica*
- C. *Naegleria fowleri*
- D. *Iodamoeba butschlii*

10. Which finding most strongly supports pathogenic *Entamoeba histolytica* rather than morphologically identical *E. dispar*?

- A. Trophozoites containing ingested RBCs
- B. Four nuclei in a cyst
- C. A central karyosome alone
- D. Presence in formed stool

11. A child develops rapidly progressive meningoencephalitis after swimming in warm freshwater. Motile amoebae are found in CSF. Which parasite is most likely?

- A. *Naegleria fowleri*
- B. *Endolimax nana*
- C. *Dientamoeba fragilis*
- D. *Entamoeba coli*

12. Which morphology favors the rhabditiform larva of *Strongyloides stercoralis* over hookworm?

- A. Short buccal cavity and prominent genital primordium
- B. Long buccal cavity and tiny genital primordium
- C. Terminal spine
- D. Operculum

13. A child develops visceral larva migrans after ingesting dog ascarid eggs. Which diagnostic approach is most appropriate?

- A. Serologic testing for *Toxocara*
- B. Skin snip
- C. Routine urine sediment
- D. Cellophane tape preparation

14. Which parasite can maintain infection through internal autoinfection and become especially dangerous during immunosuppression?

- A. *Strongyloides stercoralis*
- B. *Trichuris trichiura*
- C. *Paragonimus westermani*
- D. *Fasciola hepatica*

15. Which parasite is most appropriately investigated by examining urine for characteristic eggs?

- A. *Schistosoma haematobium*
- B. *Schistosoma mansoni*
- C. *Onchocerca volvulus*
- D. *Enterobius vermicularis*

16. A patient has chronic cough and hemoptysis from a parasitic lung infection. Eggs may be recovered from sputum. Which parasite is most likely?

- A. *Paragonimus westermani*
- B. *Trichinella spiralis*
- C. *Wuchereria bancrofti*
- D. *Hymenolepis nana*

17. Which two parasites may both be acquired from poorly cooked pork?

- A. *Taenia solium* and *Trichinella spiralis*
- B. *Taenia saginata* and *Ascaris lumbricoides*
- C. *Hymenolepis nana* and *Trichuris trichiura*
- D. *Diphyllobothrium latum* and *Toxocara canis*

18. Which human cestode produces an operculated egg?

- A. *Diphyllobothrium latum*
- B. *Taenia solium*
- C. *Hymenolepis nana*
- D. *Dipylidium caninum*

19. Where are adult *Echinococcus granulosus* tapeworms normally found?

- A. Dog intestine
- B. Human intestine
- C. Sheep liver
- D. Human blood

20. Why can humans develop either intestinal taeniasis or tissue cysticercosis from *Taenia solium*?

- A. Humans can function as definitive or accidental intermediate hosts
- B. The parasite has no intermediate host
- C. Only larvae infect humans
- D. Only eggs infect humans

21. Which cestode can complete both intermediate and definitive host stages within a human host?

- A. *Hymenolepis nana*
- B. *Taenia saginata*
- C. *Echinococcus granulosus*
- D. *Diphyllobothrium latum*

22. In a blood smear, *Babesia* may be confused with the early ring forms of which malaria parasite?

- A. *Plasmodium falciparum*
- B. *Plasmodium vivax*
- C. *Plasmodium ovale*
- D. *Plasmodium malariae*

23. Which stain is particularly useful for demonstrating *Cryptosporidium* oocysts in stool?

- A. Modified acid-fast stain
- B. Routine Gram stain
- C. India ink
- D. Wright stain

24. Which filarial parasite is normally diagnosed from skin snips rather than peripheral blood?

- A. *Onchocerca volvulus*
- B. *Wuchereria bancrofti*
- C. *Brugia malayi*
- D. *Loa loa*

25. Which malaria species is associated with the most severe disease, including cerebral involvement and blackwater fever?

- A. *Plasmodium falciparum*
- B. *Plasmodium vivax*
- C. *Plasmodium ovale*
- D. *Plasmodium malariae*

26. A daycare center has an outbreak of diarrhea. Which group contains parasites specifically highlighted as important possibilities?

- A. *Giardia*, *Dientamoeba*, and *Cryptosporidium*
- B. *Entamoeba coli*, *Endolimax nana*, and *Trichomonas vaginalis*
- C. *Toxoplasma*, *Schistosoma*, and *Taenia*
- D. *Onchocerca*, *Loa loa*, and *Wuchereria*

27. What is the principal purpose of a formalin-ethyl acetate concentration procedure in stool parasitology?

- A. Concentrating protozoan cysts and helminth eggs
- B. Demonstrating trophozoite motility
- C. Culturing malaria parasites
- D. Recovering microfilariae from blood

28. Which feature is characteristic of an *Iodamoeba butschlii* cyst?

- A. Large glycogen vacuole
- B. Terminal spine
- C. Multiple polar filaments
- D. Ingested RBCs

29. Which test can be used to assess whether *Schistosoma* eggs remain viable?

- A. Miracidial hatching test
- B. Germ tube test
- C. Knott concentration
- D. KOH preparation

30. Which two parasites should be strongly considered during a waterborne diarrheal outbreak?

- A. Giardia and Cryptosporidium
- B. Trichomonas vaginalis and Toxoplasma
- C. Onchocerca and Wuchereria
- D. Taenia and Trichinella

31. Which organisms are commonly detected using fecal antigen immunoassays?

- A. Giardia and Cryptosporidium
- B. Ascaris and Trichuris
- C. Strongyloides and hookworm
- D. Taenia and Echinococcus

32. Microsporidial infection most often begins at which body site according to the source?

- A. Gastrointestinal tract
- B. Central nervous system
- C. Bone marrow
- D. Cardiac muscle

33. Acanthamoeba keratitis has most commonly been linked to what exposure in the source?

- A. Contaminated contact-lens care solutions
- B. Eating undercooked pork
- C. Mosquito bites
- D. Freshwater fish

34. Which culture approach is highly sensitive for recovering Acanthamoeba from corneal material or lens solution?

- A. Non-nutrient agar seeded with E. coli
- B. Sabouraud agar alone
- C. Blood agar in CO₂
- D. Thioglycollate broth

35. Microsporidia have been particularly important opportunistic pathogens in which population?

- A. Immunocompromised patients
- B. Only healthy adults
- C. Only neonates
- D. Only travelers

36. How does the decolorization step of a modified acid-fast stain for coccidia differ from routine acid-fast bacillus staining?

- A. A weaker acid decolorizer is used
- B. No decolorizer is used
- C. Only alcohol is used
- D. A stronger acid-alcohol mixture is required

37. Which symptom-parasite combination is most appropriate?

- A. Giardiasis - malabsorption
- B. Trichuriasis - prominent myalgia
- C. Enterobiasis - dysentery
- D. Onchocerciasis - hematuria

38. Which egg is characterized by a thin shell, a six-hooked embryo, and polar filaments?

- A. Hymenolepis nana
- B. Taenia species
- C. Trichuris trichiura
- D. Schistosoma mansoni

39. What is the Knott concentration technique designed to recover?

- A. Microfilariae from blood
- B. Operculated eggs from stool
- C. Strongyloides larvae from stool
- D. Amoebic trophozoites from CSF

40. Which Schistosoma species has an egg with a prominent lateral spine?

- A. Schistosoma mansoni
- B. Schistosoma haematobium
- C. Schistosoma japonicum
- D. Schistosoma mekongi

41. Which parasitology request should be treated as an urgent STAT examination?

- A. Blood films for malaria
- B. Routine pinworm tape examination
- C. Routine Giardia O&P;
- D. Hydatid cyst serology

42. An immunosuppressed patient develops recurrent pneumonia, abdominal symptoms, gram-negative sepsis, and has a remote history of residence in Southeast Asia. Which chronic parasitic infection should be suspected?

- A. *Strongyloides stercoralis*
- B. *Naegleria fowleri*
- C. *Trypanosoma cruzi*
- D. *Paragonimus westermani*

43. Eosinophilic meningoencephalitis caused by a rat lungworm is associated with which parasite?

- A. *Angiostrongylus cantonensis*
- B. *Necator americanus*
- C. *Strongyloides stercoralis*
- D. *Ancylostoma braziliense*

44. In hydatid disease, how is the human host classified?

- A. Accidental intermediate host
- B. Definitive host
- C. Vector
- D. Reservoir definitive host

45. A hunter develops fever, muscle pain, and periorbital edema after eating undercooked bear meat. Which parasite is most likely?

- A. *Trichinella spiralis*
- B. *Taenia saginata*
- C. *Toxoplasma gondii*
- D. *Hymenolepis nana*

46. Accidental ingestion of *Taenia solium* eggs places humans in which role and produces what disease?

- A. Intermediate host with cysticercosis
- B. Definitive host with hydatid disease
- C. Vector with taeniasis
- D. Reservoir with malaria

47. A transplant recipient receiving immunosuppressive therapy develops persistent diarrhea. Which organism-method pairing is especially appropriate?

- A. Microsporidia - modified trichrome stain
- B. *Toxoplasma* - Gram stain
- C. *Trichinella* - trichrome stool stain
- D. *Paragonimus* - urine sediment

48. A traveler returning from India has eosinophilia, lymphadenopathy, and swelling involving the spermatic cord. Which laboratory approach is most appropriate for suspected filariasis?

- A. Examine thick blood films for microfilariae
- B. Perform a pinworm tape test
- C. Perform a KOH skin scraping
- D. Examine sputum for operculated eggs

49. Which coccidian parasite is particularly important for enteric precautions because its oocysts are immediately infective when passed in stool?

- A. *Cryptosporidium*
- B. *Cystoisospora belli*
- C. *Cyclospora cayetanensis*
- D. *Toxoplasma gondii*

50. A patient from Brazil has chronic cardiomyopathy. Myocardial tissue contains numerous amastigotes. Which parasite is the most likely cause?

- A. *Trypanosoma cruzi*
- B. *Trypanosoma brucei*
- C. *Leishmania donovani*
- D. *Babesia microti*

51. Which history is especially important when evaluating blood films for malaria?

- A. Travel location, return date, and antimalarial drug history
- B. Favorite foods only
- C. Vaccination history only
- D. Family history of anemia only

52. A municipal water system is linked to a large diarrheal outbreak. Which parasite is particularly well documented in this setting?

- A. *Cryptosporidium*
- B. *Cystoisospora belli*
- C. *Dientamoeba fragilis*
- D. *Schistosoma mansoni*

53. Outbreaks associated with imported berries, basil, leafy greens, and similar fresh produce should raise suspicion for which coccidian parasite?

- A. *Cyclospora cayetanensis*
- B. *Cystoisospora belli*
- C. *Cryptosporidium* only
- D. *Toxoplasma gondii*

54. What is the preferred diagnostic specimen for *Onchocerca volvulus* microfilariae?

- A. Skin snip
- B. Peripheral blood at midnight
- C. Urine sediment
- D. Sputum

55. Which helminth egg has a barrel-like shape with polar plugs at both ends?

- A. *Trichuris trichiura*
- B. Hookworm
- C. *Fasciola hepatica*
- D. *Diphyllobothrium latum*

56. Ingestion of embryonated eggs from which animal ascarid can produce visceral or ocular larva migrans in humans?

- A. *Toxocara canis*
- B. *Strongyloides stercoralis*
- C. *Schistosoma japonicum*
- D. *Opisthorchis sinensis*

57. Which feature is shared by *Plasmodium vivax* and *Plasmodium ovale*?

- A. Schuffner dots and a true relapse stage
- B. No malarial pigment
- C. Preference for old RBCs
- D. Typical multiple applique forms

58. Which pair of parasites can undergo internal autoinfection?

- A. *Cryptosporidium* and *Strongyloides stercoralis*
- B. *Giardia* and *Entamoeba coli*
- C. *Cystoisospora* and *Giardia*
- D. *Toxoplasma* and *Trichuris*

59. An immunocompromised patient has chronic diarrhea despite repeated routine O&P; examinations being negative. Which organisms may require special stains and remain undetected on routine studies?

- A. *Cryptosporidium* and microsporidia
- B. *Taenia solium* and *Entamoeba coli*
- C. *Ascaris* and *Trichuris*
- D. *Giardia* and *Balantidium* only

60. What microscopic structure can confirm that a tiny spore is microsporidial?

- A. Polar tubule
- B. Operculum
- C. Terminal spine
- D. Four nuclei

61. The early ring forms of *Plasmodium knowlesi* may closely resemble which malaria species?

- A. *Plasmodium falciparum*
- B. *Plasmodium vivax*
- C. *Plasmodium ovale*
- D. *Plasmodium malariae*

62. Which parasite stage is immediately infective when passed in human stool?

- A. *Cryptosporidium* oocyst
- B. *Ascaris* egg
- C. *Cyclospora* oocyst
- D. *Schistosoma* egg

63. Older trophozoites and schizonts of *Plasmodium knowlesi* can resemble which species?

- A. *Plasmodium malariae*
- B. *Plasmodium falciparum*
- C. *Plasmodium vivax*
- D. *Plasmodium ovale*

64. Which parasite's oocysts may show autofluorescence and can then be confirmed with modified acid-fast staining?

- A. *Cyclospora cayentanensis*
- B. *Entamoeba histolytica*
- C. *Dientamoeba fragilis*
- D. *Toxoplasma tachyzoites*

65. Which feature makes *Plasmodium knowlesi* potentially dangerous?

- A. A rapid 24-hour erythrocytic cycle and ability to infect RBCs of different ages
- B. A 72-hour cycle limited to old RBCs
- C. An inability to multiply in humans
- D. Only dormant liver infection

66. Which laboratory combination is appropriate for confirming microsporidial infection according to the source?

- A. Light microscopy with a modified trichrome stain
- B. Routine Gram stain with bright-field microscopy
- C. Modified acid-fast stain only
- D. India ink preparation

67. What is an important limitation when interpreting *Blastocystis* species microscopically?

- A. Morphologically similar subtypes may differ in pathogenicity
- B. Every visible organism is definitely pathogenic
- C. Only one subtype infects humans
- D. Organism number alone proves pathogenicity

68. Why should malaria thick and thin films be prepared promptly from EDTA blood?

- A. Delayed preparation can alter parasite morphology and reduce organism recovery
- B. EDTA immediately kills all RBCs
- C. Malaria parasites cannot be stained from fresh blood
- D. Only serum can preserve parasites

69. What is a defining preparation feature of a malaria thick blood film?

- A. RBCs are lysed during or before staining
- B. RBC morphology is preserved by fixation
- C. Less blood is examined than in a thin film
- D. It is primarily used for species-level RBC morphology

Answer Key & Rationales

Check this section after completing the questions.

1. B

Rationale: *Enterobius vermicularis* eggs are deposited around the perianal region, so serial adhesive-tape preparations are preferred over routine stool examination.

2. A

Rationale: Charcot-Leyden crystals are associated with breakdown products of eosinophils and may be encountered in parasitic disease.

3. A

Rationale: *D. latum* infection is acquired through ingestion of infected raw or inadequately cooked freshwater fish.

4. A

Rationale: *T. brucei gambiense* and *T. brucei rhodesiense* are transmitted through the bite of the tsetse fly.

5. A

Rationale: *Cystoisospora belli* produces oocysts that can be demonstrated in stool using modified acid-fast staining.

6. D

Rationale: *T. vaginalis* is a urogenital parasite, so fecal material is not the expected diagnostic specimen.

7. A

Rationale: A trichrome-stained fecal smear is a preferred method for demonstrating *D. fragilis* trophozoites.

8. A

Rationale: *Giardia* cysts can contaminate food and water and have caused well-described waterborne outbreaks.

9. A

Rationale: *E. gingivalis* inhabits the mouth and may ingest polymorphonuclear leukocytes.

10. A

Rationale: Ingested RBCs within trophozoites are an important morphologic clue supporting true pathogenic *E. histolytica*.

11. A

Rationale: *N. fowleri* is a free-living amoeba associated with warm freshwater and causes primary amebic meningoencephalitis.

12. A

Rationale: The *Strongyloides rhabditiform* larva has a short buccal cavity and a relatively prominent genital primordium.

13. A

Rationale: *Toxocara* species cause visceral or ocular larva migrans in humans, and serology is the diagnostic method emphasized in the source.

14. A

Rationale: *S. stercoralis* can undergo internal autoinfection, permitting infection to persist for years and progress to hyperinfection in immunosuppressed patients.

15. A

Rationale: Adult *S. haematobium* inhabit veins around the bladder, and their eggs are commonly recovered from urine.

16. A

Rationale: Adult *P. westermani* inhabit the lung, allowing eggs to be coughed into sputum and sometimes swallowed into stool.

17. A

Rationale: *T. solium* cysticerci and encysted *T. spiralis* larvae can both be acquired by eating inadequately cooked infected pork.

18. A

Rationale: *D. latum* is the operculated cestode emphasized in the source.

19. A

Rationale: Dogs are the definitive hosts and harbor adult *E. granulosus*; humans may become accidental intermediate hosts with hydatid cysts.

20. A

Rationale: Eating cysticerci can lead to an adult intestinal worm, whereas ingesting eggs can lead to tissue cysticerci, making humans either definitive or accidental intermediate hosts.

21. A

Rationale: *H. nana* can complete its life cycle in one human, with tissue cysticercoid and adult intestinal stages.

22. A

Rationale: *Babesia* often resembles *P. falciparum* ring forms and may show multiple rings within erythrocytes.

23. A

Rationale: *Cryptosporidium* oocysts can be visualized as small acid-fast-positive structures using modified acid-fast staining.

24. A

Rationale: *O. volvulus* microfilariae are located mainly in superficial skin tissues, so skin snips are the appropriate diagnostic specimen.

25. A

Rationale: *P. falciparum* can produce severe hemolysis, multiorgan disease, blackwater fever, and cerebral malaria.

26. A

Rationale: *Giardia*, *Dientamoeba fragilis*, and *Cryptosporidium* have all been implicated in nursery-school outbreaks.

27. A

Rationale: The concentration procedure improves recovery of protozoan cysts, coccidian oocysts, and helminth eggs from feces.

28. A

Rationale: *I. butschlii* cysts characteristically contain a large glycogen vacuole, readily demonstrated with iodine.

29. A

Rationale: Viable *Schistosoma* eggs can hatch and release miracidia, making the miracidial hatching procedure useful for viability assessment.

30. A

Rationale: Both *Giardia* cysts and *Cryptosporidium* oocysts are transmitted through contaminated water and have caused outbreaks.

31. A

Rationale: Fecal immunoassays are widely used to detect *Giardia* and *Cryptosporidium* antigens.

32. A

Rationale: The source describes the gastrointestinal tract as the likely initial site after ingestion of infective microsporidial spores.

33. A

Rationale: Many *Acanthamoeba* eye infections have been associated with contaminated lens-care solutions, especially homemade saline.

34. A

Rationale: *Acanthamoeba* can be recovered on non-nutrient agar seeded with *E. coli*, which serves as a food source.

35. A

Rationale: Human microsporidiosis has been most strongly associated with immunocompromised patients, including those with AIDS.

36. A

Rationale: Modified acid-fast procedures commonly use a weaker sulfuric-acid decolorizer than the acid-alcohol used for mycobacteria.

37. A

Rationale: Giardiasis may cause malabsorption. The other pairings do not match the clinical associations emphasized in the source.

38. A

Rationale: *H. nana* eggs have a thin shell surrounding an oncosphere and characteristic polar filaments.

39. A

Rationale: Knott concentration lyses red cells and concentrates microfilariae from blood for microscopic examination.

40. A

Rationale: *S. mansoni* eggs have a large lateral spine, while *S. haematobium* eggs characteristically have a terminal spine.

41. A

Rationale: Malaria blood films require rapid processing and examination because delayed diagnosis, especially with *P. falciparum*, can be fatal.

42. A

Rationale: Latent *Strongyloides* infection can persist through autoinfection and progress to hyperinfection or disseminated disease during immunosuppression.

43. A

Rationale: *A. cantonensis* is associated with eosinophilic meningitis/meningoencephalitis and larval migration.

44. A

Rationale: Humans accidentally ingest *E. granulosus* eggs and develop hydatid cysts, functioning as accidental intermediate hosts.

45. A

Rationale: *Trichinella* larvae can be acquired from inadequately cooked bear meat and cause fever, myalgia, and periorbital edema.

46. A

Rationale: Ingested *T. solium* eggs release larvae that migrate into human tissues, producing cysticercosis and making the human an accidental intermediate host.

47. A

Rationale: Microsporidia can cause severe diarrhea in immunocompromised patients, and modified trichrome staining is a recommended diagnostic method.

48. A

Rationale: The clinical setting suggests lymphatic filariasis, for which thick blood films can aid recovery of circulating microfilariae.

49. A

Rationale: *Cryptosporidium* oocysts are immediately infective on passage and can therefore contribute to nosocomial transmission.

50. A

Rationale: *T. cruzi* causes Chagas disease and can persist as amastigotes in cardiac and other striated muscle.

51. A

Rationale: Travel destination, timing of return, and antimalarial use can strongly affect species likelihood, relapse considerations, drug resistance, and parasite density.

52. A

Rationale: *Cryptosporidium* has caused well-documented outbreaks associated with contaminated municipal water.

53. A

Rationale: *Cyclospora* has been linked to foodborne outbreaks involving fresh produce such as berries, basil, and leafy vegetables.

54. A

Rationale: *Onchocerca microfilariae* are found in superficial skin rather than routinely circulating in peripheral blood.

55. A

Rationale: *T. trichiura* eggs have a characteristic thick shell and bipolar plugs.

56. A

Rationale: Humans can ingest infective *Toxocara* eggs and become accidental hosts for migrating larvae.

57. A

Rationale: Both *P. vivax* and *P. ovale* infect younger RBCs, may display Schuffner stippling, and can relapse because of persistent liver stages.

58. A

Rationale: The source specifically notes internal autoinfection in *Cryptosporidium* and *S. stercoralis*.

59. A

Rationale: *Cryptosporidium* requires modified acid-fast methods and microsporidia require specialized modified trichrome staining, so routine O&P; methods may miss them.

60. A

Rationale: Demonstration of the characteristic polar tubule supports confirmation of microsporidial spores.

61. A

Rationale: Early *P. knowlesi* rings can resemble *P. falciparum*, creating a potential identification problem.

62. A

Rationale: *Cryptosporidium* oocysts are already infective when excreted, unlike several other parasites that require environmental development.

63. A

Rationale: More mature *P. knowlesi* stages can resemble *P. malariae*, including band-like forms.

64. A

Rationale: *Cyclospora* oocysts exhibit autofluorescence, a useful presumptive feature before confirmatory staining.

65. A

Rationale: *P. knowlesi* has a rapid 24-hour erythrocytic cycle, can involve RBCs across age groups, and may cause serious disease.

66. A

Rationale: Modified trichrome staining examined by light microscopy is a recommended approach for demonstrating microsporidial spores.

67. A

Rationale: The source notes multiple morphologically indistinguishable subtypes or strains, some potentially pathogenic and others nonpathogenic.

68. A

Rationale: Holding EDTA blood too long can change parasite morphology, reduce organism recovery, and impair staining quality.

69. A

Rationale: Thick films require lysis of RBCs, allowing a larger volume of blood to be screened sensitively for parasites.